ALEC CHRISTOV

Electrical Design Engineer

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24 york Street, Eltham

EDUCATION

Bachelor of Engineering (electrical and electronic engineering) (honours)/Bachelor of Business(management)

RMIT University 2020-2026

Victorian Certificate of Education

Certificate III In Applied Language (Mandarin)

Certificate III In Hospitality Back Of House Operations

Eltham College 2015-2020

TECHNICAL SKILLS

- Electrical design and schematic capture
- PCB layout and design iteration
- Component selection and datasheet interpretation
- BOM creation, management, and procurement
- Prototype assembly, testing, and debugging
- Autodesk Fusion 360 Electronics
- Working knowledge of Altium Designer

About Me

I am an enthusiastic graduate electrical engineer with hands-on experience in end-to-end electrical design and product development. I have an inquisitive and tenacious approach to solving problems. I thrive in a team environment and have excellent people skills, and I take every opportunity to learn when I find great mentors and professionals who can guide me and help me develop.

WORK EXPERIENCE

March 2021 - Present

Unison Technology

Electrical Design Engineer

Contributed extensively to the design and development of a specialised consumer bicycle safety product within a small team, taking significant responsibility across the end-to-end electrical design process. Following initial mentorship from a Senior Engineer and IT Architect, progressed into a lead role across electrical design, prototype testing, and product development. Responsibilities included interpreting functional requirements, translating concepts into schematic designs, developing PCB layouts, selecting components, managing BOMs and procurement, assembling and testing prototypes, and refining designs through multiple iterations with increasing focus on manufacturability. In this role, I have:

- Translated product concepts and functional requirements into complete schematic designs
- Developed 2D PCB layouts and used 3D modelling to verify spatial and mechanical fit
- Selected components and created BOMs with consideration for availability and obsolescence
- Managed component sourcing, procurement, and PCB manufacturer coordination
- Conducted manual SMT assembly, prototype bring-up, and hardware/software verification
- Supported iterative design refinement to improve functionality, reliability, and manufacturability
- Researched manufacturing pathways, production methods, and commercial rollout considerations

Current project stage: Progressing the third design iteration with emphasis on manufacturability, production readiness, and small-scale manufacturing implementation.

PROFESSIONAL SKILLS

- Leadership and team coordination
- Mentoring and training
- Communication and stakeholder management
- Problem-solving and initiative
- Composure under pressure

EXTRA CURRICULAR

Volunteer Youth Soccer Coach Program

Coached in a youth soccer program for 2 years, coaching ages 5-7 on Saturday mornings.

Hobby Electronics Projects

- Designed and built an electric bike with a custom-built battery
- Standalone three-port PV system with Wi-Fi-enabled power management and sensing, managed by a custom PCB
- Custom Lithium Ion charging circuit boards
- DIY microwave transformer metal melting rig
- Current - Designing Dual-Resonance Steady-State Tesla Coil

Feb 2021 to March 2021
Wallis Social Research

Interviewer

Social research interviewer – WFH casual call-assisted-telephone interviewing. This was a short-term role I did part-time while studying. I developed cold-calling skills and confidence on the phone, as we were expected to entice people to complete surveys for State and Federal Social Research projects.

2022 - Present
Bulleen/Templestowe Basketball Association and Basketball Victoria

Accredited Referee Coach

Acted in an authoritative capacity, responsible for the instruction, mentorship, and formal accreditation of basketball referees.
Attained Grade 1 accreditation in 2022, authorising certification of C-grade referees and assessment of B-grade referees.
Promoted to Grade 2 accreditation in 2024, granting authority to accredit referees up to A-grade and formally evaluate Level 1 (Elite Junior State Level) referees. Additionally co-led introductory referee training courses, ensuring a structured, professional, and supportive environment for novice officials to confidently navigate the demands of high-pressure officiating.

2017 - 2024
Bulleen/Templestowe Basketball Association and Basketball Victoria

Referee

Basketball referee on both weeknights and weekends. This involves managing all elements of a basketball game: scoreboard management; game control; coach and spectator management; and resolving any conflicts efficiently to maintain a calm and composed environment for all involved.
I also have responsibility for mentoring assigned rookies, who I shadow as they learn, giving practical guidance and support throughout games, pre and post-game feedback, and written performance appraisals/feedback.
Included selection as representative of my basketball association as a Level 2 Panel referee for 3 years. This involved refereeing BIGV (Victorian State Basketball league) in addition to National Basketball Tournaments at an elite level.
Appointed as a State Representative Referee at national wheelchair tournaments, showcasing authoritative game management and decisive conflict resolution capabilities in professional-grade, high-intensity settings.

REFERENCES

Drew Lethbridge
Unison Technology / Senior IT Architect
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Mitchell Pratt
Bulleen Basketball / Referee Advisor
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